

IK 502: Introduction to Bio-signals

Course Code	IK 502
Course Name	Introduction to Bio-signals
L-T-P-C	3-0-2-4
Prerequisites	None
Intended for	BTech/MTech/MS/MSc/MA/Ph.D.
Mutual Exclusion	None
Approval	50th BoA (Jul 2023); 42nd Senate (Jan 2024)

Course Contents

- **Mathematical Preliminaries:** Fourier transform, sampling and filtering, Solution to wave equation in spherical co-ordinate system, Introduction of Spherical Harmonics.(3 hours)
- **Basics of bio-signals:** Definition and models of bio-signals, types of bio-signals, bio-signals monitoring, Pre-processing for bio-signals, bio-signals analysis, and classification of bio-signals. (2 hours)
- **Brain signals:** Human Brain Anatomy, Electroencephalogram (EEG) and magnetoencephalogram (MEG) signals, recording of EEG and MEG signals, EEG signals characteristics and rhythms, evoke potentials, diagnosis of central nervous systems disorders based on brain-signals, various approaches for analysis, feature extraction, and classification of brain signals, MRI and FMRI basics, BOLD signal acquisition, applications of FMRI (10 hours)
- **Brain Source Localization and connectivity:** Array Signal Processing Basics - Data model, correlation and subspace based (MUSIC) localization, Brain Source Localization: Forward & Inverse Problem, Introduction of Head harmonics for brain source localization (BSL), Application of BSL in BCI control, Epileptogenic zone detection. Brain connectivity representation, decomposition methods and types of networks, Clinical and cognitive applications of brain connectivity. (10 hours)
- **Cardiac signals:** Electrocardiogram (ECG) and phonocardiogram (PCG) signals, recording process of ECG and PCG signals, heart rate variability (HRV) signals, diagnosis of heart diseases based on cardiac signals, various methods for analysis, feature extraction, and classification for cardiac signals. (8 hours)
- **Muscle signals:** Electromyogram (EMG) signal, motor unit action potentials (MUAP), EMG and neuromuscular diseases, feature extraction of EMG, analysis and classification methods for EMG signals. (6 hours)
- **Other bio-signals:** Pulse signals, blood pressure, blood flow, photoplethysmogram, electrooculogram, electroretinogram, center of pressure, and respiratory signals. (3 hours)
- **Laboratory/practical/tutorial Modules:** The course will involve practical assignments which can be conducted in the lab, and would also involve programming assignments.

Text Books

1. R.M. Rangayyan, **Biomedical Signal Analysis: A case Based Approach**, IEEE Press, John Willy & Sons. Inc, 2002.
2. Kayvan Najarian and Robert Splinter, **Biomedical Signal and Image Processing**, 2nd Edition, CRC Press, 2005.

References

1. M.A. Jatoi and N. Kamel, **Brain source localization using EEG signal analysis**, CRC Press, 2017
2. Boaz Rafaely, **Fundamentals of spherical array processing**, Springer, 2015
3. HL Van Trees, **Optimum Array Processing**, New York: Wiley, 2002
4. Scott Heutzel, Allen Song, Gregory McCarthy, **Functional Magnetic Resonance Imaging**, 2nd Edition, Sinauer Associates, 2009 .